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The Editor-in-Chief

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Re: Regular Paper submission, “FAIR-X: A Threshold-Transfer Evaluation Contract for Behavioral Malware Detection under Distribution Shift”

Dear Editor-in-Chief,

We respectfully submit the above manuscript for consideration as a Regular Paper in *IEEE Transactions on Dependable and Secure Computing* (TDSC). The submission comprises an 18-page main manuscript and a separate four-page supplementary file. The paper addresses a concrete dependability problem in security-machine-learning systems: a detector can rank malicious processes well yet produce unreliable operational decisions when its validation-derived threshold does not transfer to a shifted deployment distribution.

Scope and relevance to TDSC. The work concerns dependable evaluation of endpoint-security detectors under class imbalance, scarce malicious validation data, and distribution shift. It is directly relevant to TDSC because it studies when a learned security component can and cannot support a reproducible alerting decision. The paper connects model evaluation to system dependability through explicit controls for test isolation, feature and label leakage, calibration, common operating policies, uncertainty, early-detection limits, host/time separation, and response boundaries.

Scientific gap and contribution. Published security-ML guidance warns against data leakage, weak baselines, and inappropriate metrics. However, these principles do not by themselves provide an executable procedure for deciding whether a hybrid detector adds value under the same operating constraints as a strong non-hybrid baseline. FAIR-X closes this gap through:

- a general threshold-transfer evaluation contract that fixes data access, feature fitting, calibration, threshold selection, baseline parity, and uncertainty reporting before official-test evaluation;
- three quantitative diagnostics—tabular dominance, score compression, and threshold sensitivity—that make fusion claims falsifiable;
- repeated cross-fitted threshold selection over all available BETH development positives, without synthetic enrichment;
- target-copy-free GRU/LSTM capacity ablations that bound the sequence-model conclusion; and
- an instantiation on BETH and independent protocol validation on MalBehavD-V1, supplemented by BETH prefix, calibration, attribution, cost, feature-stress, host-disjoint, and temporal audits.

Principal evidence. After process aggregation, the official BETH training and validation files contain no malicious processes, whereas 121 of 198 official-test processes are malicious. Under one common validation-FPR policy, TabRF obtains an F1 score of 0.7210, RF-500 obtains 0.7043, and MetaRF obtains 0.6756; the hybrid fusion therefore provides no evidence of improvement. Fifty thresholds fixed by repeated five-fold validation have median 0.006 and IQR 0.004–0.006. Yet the locked median threshold converts a 5% development-FPR objective into 32.47% test FPR, with F1 0.6964. This separates threshold sampling stability from threshold transfer and is the paper’s main dependability finding.

The ranking–decision distinction is independently visible in ExtraTrees, which achieves the best BETH ranking (AUC 0.7973, AP 0.8572) but only F1 0.6180 at its locked operating point. Target-copy-free recurrent models remain weak across 23 666–241 074 parameters (AUC 0.4160–0.4911), so the paper limits this negative result to the tested objectives and representation. On MalBehavD-V1, the same evaluation contract yields repeated-split median F1 0.9533, supporting protocol portability without claiming direct cross-schema model transfer.

Claim boundaries and reproducibility. The manuscript does not claim ransomware-family attribution, calibrated posterior risk, direct BETH-to-MalBehavD model transfer, or readiness for autonomous containment. These exclusions follow from the available labels, schemas, test size, and early-detection results. The reproducibility package, including executable scripts, generated CSV/JSON artifacts, the main manuscript, and separate supplementary material, is publicly available at github.com/EkodeckStephane/fair-beth-malicious-process-detection. The submitted source archive includes the bibliography, IEEE class and bibliography-style files, and every active figure required to compile the main manuscript.

Mandatory author declarations.

- **Originality and exclusive submission:** The manuscript is original, has not been published, is not under consideration elsewhere, and will not be submitted to another venue while under TDSC review.
- **Author approval:** All named authors have read and approved the manuscript, its authorship order, and the stated contributions.
- **Conflicts and funding:** The authors declare no competing interests. The research received no external funding.
- **Ethics and data:** The study involves no human participants, animals, or sensitive personal data. It uses publicly available cybersecurity datasets and reports derived experimental artifacts.
- **Supplementary material:** Detailed feature specifications, hyperparameters, threshold sensitivity, data/model cards, deployment boundaries, and audit figures are supplied as a separate four-page file, with independent Table S1–S3 and Figure S1–S4 numbering, consistent with TDSC submission requirements.

We believe the manuscript will interest TDSC readers working on dependable security analytics, evaluation under distribution shift, endpoint detection, and reproducible security-machine-learning methodology. Thank you for your consideration.

Yours sincerely,

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On behalf of Vivien Beyala Kamgang, Serge Alain Ebélé, Ulrich Joel Atchiengang Nguéyep, Stéphane Willy Mossebo Tcheunteu, Damaris Belle M. Fotso, Arthur Ulrich Ewane, and Julius Marcellin Nkenlifack